



Explainable AI for SMS spam filtering: A novel hybrid architecture combining fuzzy logic and bidirectional LSTM networks

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ABSTRACT

SMS spam filtering has become a critical application in modern telecommunications, where unsolicited and fraudulent text messages compromise user security, erode trust, and burden network resources. To address these challenges, this study proposes a novel hybrid architecture that integrates fuzzy logic with a bidirectional long short-term memory (BiLSTM) network to achieve high classification performance and transparent decision-making. The fuzzy logic module captures linguistic uncertainty using Gaussian membership functions, while the BiLSTM effectively models sequential dependencies in text. To further enhance interpretability, token-level attention highlights influential words, and rule-based explanations provide human-understandable justifications for each classification outcome. The proposed framework was evaluated on two benchmark datasets: the UCI SMS Spam Collection and the ExAIS_SMS dataset. Experimental results demonstrate strong performance, with accuracies of 99.4 % and 95.8 %, respectively, outperforming several baseline approaches. Notably, the model also achieves high discriminative capability, with AUC values of 0.994 and 0.980 across the two datasets. Beyond accuracy, the framework is computationally efficient, requiring <2.5 GB of memory and averaging under 7 ms per message during inference on a consumer-grade GPU. This ensures feasibility for real-time, large-scale deployment. By balancing accuracy, interpretability, and efficiency, the proposed fuzzy-deep hybrid approach offers a practical and trustworthy solution for SMS spam detection in telecom environments.

1. Introduction

Short Message Service (SMS) remains a vital communication channel in telecommunications, despite the dominance of instant messaging platforms. Its ubiquity, simplicity, and cross-device compatibility make it indispensable for personal and professional communication [1]. However, this widespread use also exposes SMS to exploitation by malicious actors who disseminate spam, phishing links, and fraudulent offers [2]. Such messages degrade user experience and create significant financial and cybersecurity risks, especially in regions where mobile connectivity is the primary digital access channel. For instance, global traffic reached nearly 8.3 trillion SMS messages annually in 2017, underscoring the scale of the problem and the urgent need for reliable detection mechanisms [3]. More recent statistics illustrate the accelerating severity of this threat. In the first half of 2023, users in the United States received approximately 78 billion spam text messages, while in February 2025 alone, they received nearly 19.2 billion unsolicited SMS

messages [4]. These alarming figures confirm that SMS spam continues to grow in volume and sophistication, making the development of scalable, accurate, and interpretable detection systems an urgent research priority [5]. Despite notable advances in machine learning and deep learning for SMS spam detection, a clear gap persists between high accuracy and transparent decision-making. Deep models often operate as black boxes, limiting their suitability for regulated telecom environments, whereas fuzzy-rule systems offer interpretability but cannot capture sequential and contextual dependencies. This gap underscores the need for a unified framework that integrates uncertainty modeling, contextual sequence learning, and explainable reasoning within a scalable architecture. Early spam detection research relied on machine learning (ML) approaches such as Naive Bayes, Support Vector Machines (SVM), and Decision Trees, achieving moderate success with hand-crafted features and word-frequency representations. Nevertheless, these models struggle with linguistic ambiguity, adversarial obfuscation (e.g., “cl1ck h3re”), and limited cross-domain generalization [6]. More

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recently, deep learning (DL) techniques, including Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) such as Long Short-Term Memory (LSTM) and Bidirectional LSTM (BiLSTM), have advanced the state of the art through automatic feature extraction and sequence modeling [7]. Yet, these methods often behave as black boxes: they achieve strong accuracy but fail to provide transparent reasoning, a limitation that undermines trust and hinders deployment in sensitive telecom applications [8].

1.1. Motivation and novelties

The growing sophistication of SMS spam, characterized by obfuscated tokens, deceptive linguistic patterns, and increasingly adaptive phishing strategies, has exposed critical limitations in both conventional machine learning and modern deep learning approaches. Traditional fuzzy-rule systems offer interpretability but lack the capacity to capture sequential dependencies. At the same time, deep neural models achieve strong accuracy but operate as opaque black boxes, hindering their adoption in regulated, high-stakes environments such as telecommunications. These gaps highlight the need for a unified framework that jointly addresses uncertainty modeling, contextual feature extraction, and transparent decision-making. Building on this need, the proposed approach introduces several methodological novelties that advance the state of the art in SMS spam detection:

- **Fuzzy-Deep Hybrid Architecture:** A unified model that seamlessly integrates symbolic fuzzy reasoning with BiLSTM-based sequence modeling. This fusion enables simultaneous handling of linguistic ambiguity and deep contextual structure, achieving a balance rarely explored in prior work.
- **Fuzzy-Aware Preprocessing and Membership Design:** A novel preprocessing strategy grounded in Gaussian membership functions, designed to robustly capture obfuscated patterns (e.g., “cl1ck”, “w1n”) and encode linguistic cues such as urgency, promotion, monetary incentive, and URL suspicion. This produces uncertainty-aware representations that are more semantically informative than traditional crisp features.
- **Integrated Dual-Level Explainability (XAI Module):** An interpretability mechanism combining token-level attention from the BiLSTM with dynamic fuzzy IF-THEN rules. This dual-layer explanation provides transparent, human-understandable justifications for each classification outcome, addressing key regulatory and practical requirements for auditability.
- **Comprehensive and Cross-Domain Validation:** Extensive evaluation on both the UCI SMS Spam dataset and the linguistically diverse ExAIS_SMS corpus demonstrates consistent improvements in accuracy and AUC, coupled with low computational overhead. This confirms the scalability and deployment feasibility of the proposed framework in real-world telecom environments.

1.2. Paper organization

The remainder of this paper is structured as follows: [Section 2](#) reviews related work. [Section 3](#) details the proposed methodology. [Section 4](#) introduces the datasets and experimental setup. [Section 5](#) presents the results and analysis, including computational complexity and explainability. [Section 6](#) concludes the paper.

2. Related work

The field of SMS spam detection has witnessed extensive research over the past two decades, evolving from rule-based and classical machine learning approaches to more sophisticated deep learning models [9]. To highlight this progression and the strengths and limitations of each category, we organize the literature into two main groups: Traditional Machine Learning (ML) methods and Deep Learning (DL)

approaches. This structure allows us to contrast the historical reliance on manual feature engineering and algorithmic simplicity with the modern emphasis on automated representation learning and model scalability.

2.1. Traditional machine learning approaches for SMS spam detection

Early efforts in SMS spam detection were rooted in rule-based systems [10], which later gave way to machine learning techniques. Popular ML algorithms in this domain include SVM and Random Forest (RF). These models typically depend on manual feature engineering, which is time-intensive and susceptible to dataset-specific bias. Paper [11] employed a combination of RF with Term Frequency-Inverse Document Frequency (TF-IDF) feature representation, achieving an accuracy exceeding 97 % on the UCI dataset. However, their approach relied heavily on the quality of feature extraction, limiting its adaptability to other datasets. Similarly, Saeed [12] utilized several ML models, including J48, K-Nearest Neighbors (KNN), and Decision Trees, with DT yielding the best accuracy of 97.05 % after preprocessing. In another contribution, Ali et al. [13] extracted seven handcrafted features and applied Multiple Linear Regression (MLR) and Extreme Learning Machine (ELM) classifiers, achieving 98.2 % accuracy. Despite their effectiveness, such models generalize poorly across different domains due to their handcrafted nature. Hameed [14] explored binary particle swarm optimization (PSO) for optimizing fuzzy rule selection and achieved 98.5 % accuracy using six features. However, this method faced challenges related to the complexity and scalability of fuzzy systems, especially when applied to large-scale or multilingual datasets. A recent comparative study [15] on the UCI and ExAIS_SMS datasets showed that the Decision Tree model achieved the highest accuracy of 96 % on both datasets. Naïve Bayes followed with 95 % accuracy on UCI and 85 % on ExAIS_SMS. In contrast, J48 (C4.5) performed well on UCI (95 % accuracy) but dropped to 80 % on ExAIS_SMS, highlighting challenges in handling linguistically diverse SMS data.

2.2. Deep learning-based approaches

Deep learning models have demonstrated superior performance across many fields of life applications, such as education [16] and healthcare [17], as well as text classification. In recent years, SMS spam detection tasks have benefited mainly from the ability to learn features from raw text without extensive manual preprocessing automatically. Sheikhi et al. [18] introduced a two-stage framework that combines feature extraction and classification using an averaged neural network, achieving 98.8 % accuracy. Derakhshi et al. [19] developed a model that processes SMS content using parallel CNN and LSTM networks, extracting spatial and temporal features simultaneously. Another hybrid model [20] combining CNN and GRU was proposed for SMS spam detection using Turkish and English datasets, including the UCI SMS Spam Collection. Despite its high memory requirements, the model achieved 99.07 % accuracy. Liu et al. [21] introduced a modified Transformer-based model for SMS spam detection and evaluated it on both the SMS Spam Collection v.1 and UtkML’s Twitter Spam Detection datasets. Compared to traditional classifiers and state-of-the-art approaches, the model achieved the best performance with 98.92 % accuracy, 0.9451 recall, and an F1-score of 0.9613. Its strong results on both datasets demonstrate its potential for adaptation to similar short-text classification tasks. On the ExAIS_SMS dataset, Abayomi-Alli et al. [22] employed a BiLSTM model, achieving the best overall performance with an accuracy of 93.4 %, significantly outperforming traditional machine learning methods. Shen et al. [23] proposed a hybrid model integrating BERT embeddings with triple-graph convolutional networks (G3CN) for SMS spam detection. Their approach captured semantic, syntactic, and co-occurrence patterns, achieving 99.28 % accuracy on the UCI SMS Spam Collection and 93.78 % on the Twitter dataset, outperforming several state-of-the-art baselines. Aparna et al. [24] proposed a hybrid ALBERT-BiLSTM model enhanced with

cross-attention and progressive knowledge distillation for SMS spam classification. Their approach achieved 98 % accuracy and an F1-score of 0.98 on the SMS Spam Collection dataset. Some papers used a local or unpublished dataset, such as Arabic [25] and Persian [26]. Table 1 summarizes the related works.

3. Methodology

The proposed methodology for SMS spam detection integrates traditional preprocessing techniques, fuzzy logic reasoning, deep learning models, and explainable AI to enhance classification accuracy and interpretability. The complete architecture is illustrated in Fig. 1.

3.1. Pre-processing

Effective SMS spam detection begins with robust text pre-processing, particularly given the noisy, informal, and abbreviated nature of SMS data. The pre-processing pipeline in our architecture is meticulously designed to standardize input messages and optimize them for downstream linguistic and neural processing [19]. This step improves model

Table 1

Summary of existing SMS spam detection approaches, datasets used, and achieved accuracy.

Ref.	Techniques Used	Dataset(s)	Accuracy / Metrics	Key Features / Notes
[11]	RF + TF-IDF	UCI SMS Spam Collection	97 %	Strong with feature engineering; limited adaptability
[12]	J48, KNN, Decision Tree	UCI SMS Spam Collection	97.05 % (DT best)	Decision Tree is the highest after preprocessing
[13]	MLR + ELM with handcrafted features	UCI SMS Spam Collection	98.2 %	Based on 7 manually extracted features
[14]	Binary PSO + Fuzzy Rules	UCI SMS Spam Collection	98.5 %	Fuzzy rule optimization; scalability challenges
[15]	Comparative study (DT, NB, J48)	UCI, ExAIS_SMS	DT: 96 %, NB: 95 % (UCI), 85 % (ExAIS), J48: 95 % (UCI), 80 % (ExAIS)	Showed the limits of ML methods in diverse contexts
[18]	Two-stage Averaged Neural Network	UCI SMS Spam Collection	98.8 %	Combined feature extraction and classification
[19]	Parallel CNN + LSTM	UCI SMS Spam Collection	99.28 %	Extracts spatial and temporal features jointly
[20]	Modified Transformer	UCI SMS, Twitter (UtkML)	98.92 % (Recall 0.9451, F1 0.9613)	Robust for short-text classification tasks
[21]	CNN + GRU	UCI + Turkish	99.07 %	Multilingual; high memory requirement
[22]	BiLSTM	ExAIS_SMS	93.4 %	Best on ExAIS_SMS; outperformed classical ML
[23]	BERT + Triple-Graph Convolutional Networks (G3CN)	UCI SMS Spam, Twitter	99.28 % (UCI), 93.78 % (Twitter)	Captures semantic, syntactic, and co-occurrence patterns
[24]	ALBERT-BiLSTM + Cross-Attention + Knowledge Distill.	UCI SMS Spam Collection	98 % (F1 = 0.98)	Lightweight transformer with BiLSTM; strong interpretability

performance and enhances the interpretability of features used in the Fuzzy Input Layer.

3.1.1. Lowercasing

All SMS texts are converted to lowercase to eliminate case sensitivity. This ensures that tokens like “Free” and “free” are treated identically, improving consistency across the dataset [27].

3.1.2. Stop word removal

Common English stop words (e.g., “is”, “the”) are removed using a curated stop-word list. These words typically add noise without contributing meaningfully to spam semantics.

3.1.3. Punctuation removal

Non-alphanumeric characters such as commas, periods, exclamation marks, and symbols are stripped from the text. Since punctuation is often used for emphasis or formatting in spam messages (e.g., “WIN!!!”), This step is carefully balanced to preserve semantic cues if needed by the fuzzy layer [28].

3.1.4. Lemmatization

Words are reduced to their base or root forms using lemmatization (e.g., “winning” → “win”, “messages” → “message”). This step helps unify semantically equivalent tokens and ensures more reliable matching in fuzzy rule sets and word embeddings.

3.1.5. URL and emoji removal

URLs, email addresses, and emojis are removed using regular expressions. These elements are frequent in spam messages and may disrupt the learning of token-level semantic patterns [29]. However, a separate indicator feature capturing the presence of URLs or emojis is preserved as input to the fuzzy module, as their existence can be a strong spam signal.

3.2. Feature extraction

The proposed architecture employs a hybrid feature extraction strategy comprising a Fuzzy Input Layer (LMA) and a Deep Learning Backbone (BiLSTM) to capture semantic ambiguity and contextual dependencies effectively. This dual-branch system ensures robust spam detection, even in cases of linguistic obfuscation or subtle contextual manipulation.

3.2.1. Fuzzy input layer – linguistic membership analysis (LMA)

Traditional binary or crisp feature extraction often fails to handle the inherent fuzziness of natural language, especially in adversarial spam messages that utilize deliberate ambiguity (e.g., “cl1ck h3re”, “w1n ca\$h”) [30]. To address this, the system incorporates LMA as a fuzzy logic-based mechanism for extracting uncertainty-aware features [31].

- **Fuzzy Sets & Linguistic Variables:** Terms and tokens within the SMS are assigned to predefined fuzzy sets (e.g., promotional language, urgency, monetary incentive) using a membership function that captures the degree of belonging.
- **Granular Representation:** Instead of representing words in binary fashion (e.g., spam keyword: 0 or 1), LMA encodes them continuously [0,1], reflecting their intensity or certainty within the spam context.
- **Vagueness Modeling:** This approach efficiently models semantic fuzziness, such as partially suspicious content that doesn't fully match known spam patterns but exhibits weak indicators (e.g., partial URLs, soft promotional phrases).

These fuzzy features provide interpretable, human-understandable indicators that are later fused with deep features for final prediction.

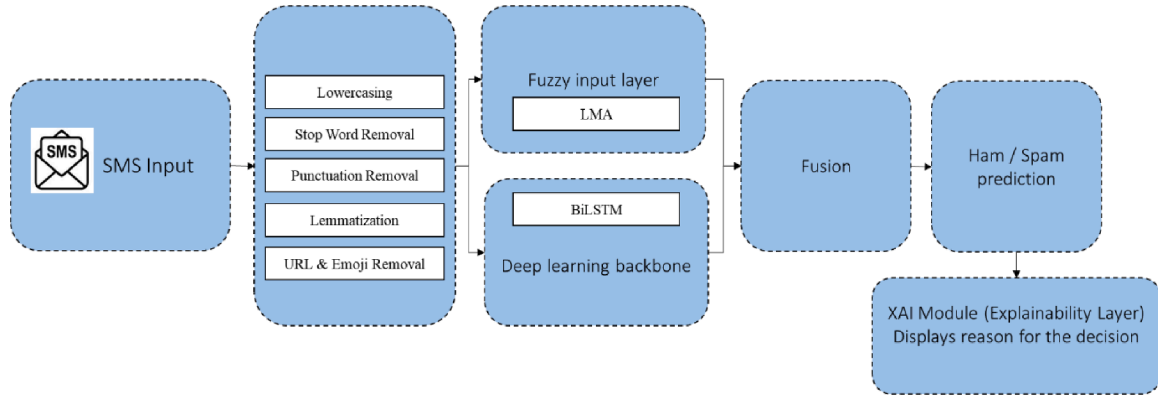


Fig. 1. Architecture of the proposed fuzzy-deep hybrid SMS spam detection system.

3.2.2. Deep learning backbone – bidirectional LSTM (BiLSTM)

While LMA captures high-level fuzzy patterns, deep contextual information is extracted using a Bidirectional Long Short-Term Memory (BiLSTM) network, a powerful variant of recurrent neural networks (RNNs) that processes sequential data in both forward and backward directions.

- **Contextual Encoding:** BiLSTM simultaneously reads the SMS text from left to right and right to left, capturing past dependencies and future context. This is essential for understanding short texts like SMS, which lack a clear structure.
- **Handling Sequential Dependencies:** Spam messages often rely on specific word orderings and patterns (e.g., “Congratulations! You’ve won...” or “Reply YES to claim...”). BiLSTM excels at modeling these temporal relationships, which are critical for accurate classification [32].
- **Embedding Layer:** Before feeding into the BiLSTM, words are mapped to pre-trained Word2Vec (100-dim, static) embeddings, preserving semantic relationships among terms. In all experiments, we adopt this configuration.
- **Regularization and Optimization:** Techniques such as dropout, batch normalization, and Adam optimization are applied to prevent over-fitting and improve generalization on unseen SMS data.

Combining fuzzy reasoning (to capture linguistic ambiguity) and deep sequential learning (to capture syntactic and contextual nuances), this hybrid feature-extraction approach enhances the model’s robustness, accuracy, and interpretability in classifying SMS messages as spam or ham.

3.3. Feature fusion and prediction

The Fuzzy Input Layer and BiLSTM outputs are combined in the Fusion Layer to form a comprehensive feature representation. We adopt a feature-level concatenation strategy [33], combining fuzzy membership scores and BiLSTM contextual embeddings into a single vector. This hybrid representation is then passed through a fully connected dense layer with a sigmoid activation function to perform binary classification (spam or ham) [34]. This fusion lets the model simultaneously consider rule-based linguistic uncertainty and deep contextual semantics.

3.4. Explainability mechanism (XAI module)

The proposed architecture incorporates a dedicated Explainability Layer to address the black-box nature of deep learning models and enhance model transparency [35]. This novel component produces human-understandable justifications for every spam or ham prediction. This layer operates at two integrated levels: linguistic rule-based

interpretation via fuzzy logic and dynamic token-level importance via deep attention mechanisms.

3.4.1. Attention-Based explanation (BiLSTM)

The model uses BiLSTM attention weights to quantify each word or token’s contribution to the final classification decision. Tokens with higher weights are those that the model deems more indicative of spam (e.g., “win”, “claim”, “free”, “urgent”). These weights are normalized and visualized in heatmaps to allow traceability and interpretability, which is especially useful for developers or compliance auditors [36]. Example output:

“High attention given to tokens: [‘urgent’, ‘click’, ‘win’].”

3.4.2. Fuzzy logic-based explanation

Simultaneously, the model leverages linguistic membership values generated by the Fuzzy Input Layer. Each term is assigned a degree of membership to one or more fuzzy sets, such as *Urgency*, *Promotion*, or *Suspicion* [37]. These scores are mapped to interpretable IF-THEN rules that mimic human-like reasoning.

Example rule:

IF “win” $\in$ High-Promotion AND URL $\in$ Suspicious-Link THEN spam-likelihood = 0.92

The rule generation is dynamic and adapts to the specific message being analyzed. Each decision includes an explanation string derived from these rules, such as:

“Classified as spam because it contains promotional language (‘win’) and a suspicious link.”

3.4.3. User-Facing justification

The output explanation is formatted as a concise, natural-language summary and is attached to the prediction result. These summaries are loggable, exportable, and auditable, making the system highly suitable for deployment in environments requiring accountability and compliance (e.g., banking, telecom, enterprise messaging) [38].

4. Evaluation

This section evaluates the proposed fuzzy-deep hybrid model using benchmark SMS datasets, focusing on accuracy, interpretability, efficiency, and robustness across diverse contexts.

4.1. Dataset description

We utilize two publicly available datasets containing labeled SMS messages, the UCI SMS Spam Collection v. 1 and the ExAIS_SMS dataset,

to evaluate the proposed fuzzy-deep hybrid model's effectiveness and robustness. These corpora represent standard benchmark data and linguistically diverse real-world message samples, enabling a comprehensive assessment of the model's performance.

4.1.1. UCI SMS spam collection dataset

The UCI SMS Spam Collection is a widely used benchmark dataset in spam detection research. It comprises 5574 SMS messages, of which 747 (13.4 %) are labeled as spam and 4827 (86.6 %) as ham (legitimate). The messages are written in English and cover a range of common spam categories, including unsolicited promotions, prize notifications, and scam alerts. This dataset is characterized by its moderate class imbalance and serves as a standard reference for evaluating SMS spam filtering models [39].

4.1.2. ExAIS SMS dataset

The ExAIS SMS dataset is a publicly available corpus that reflects SMS communication in African-English contexts. It contains 5240 SMS messages, including 2352 spam messages (44.9 %) and 2888 ham messages (55.1 %), offering a more balanced distribution than the UCI dataset. The messages in ExAIS SMS are diverse in style, with a mix of formal and informal language, localized expressions, and region-specific spam patterns. This dataset provides a vital perspective for evaluating the adaptability of spam detection models to different linguistic environments and content characteristics [40]. Fig. 2 shows the Distribution of spam and ham messages in the UCI SMS Spam Collection v.1 and ExAIS SMS datasets. Using these two datasets, one widely recognized and one more regionally contextualized, we aim to demonstrate the proposed model's applicability across benchmark and real-world SMS environments, ensuring its generalizability and practical relevance.

4.2. Evaluation metrics

To evaluate the performance of the proposed fuzzy deep learning-based SMS spam detection model, we use four commonly adopted classification metrics: Accuracy, Precision, Recall, and F1-Score. These metrics comprehensively assess how well the system distinguishes between spam and ham messages.

4.2.1. Accuracy

Accuracy measures the overall correctness of the model's predictions. It reflects the proportion of total SMS messages correctly classified as either spam or ham [41].

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} 100\% \quad (1)$$

Where:

- TP (True Positives): Spam messages correctly predicted as spam.
- TN (True Negatives): Ham messages correctly predicted as ham.
- FP (False Positives): Ham messages incorrectly predicted as spam.
- FN (False Negatives): Spam messages incorrectly predicted as ham.

Although accuracy is useful, it may not be reliable when the data is imbalanced (e.g., there are significantly more ham than spam messages). Hence, we also report other metrics.

4.2.2. Precision

Precision evaluates the reliability of spam predictions by measuring how many messages predicted as spam were, in fact, spam.

$$Precision = \frac{TP}{TP + FP} 100\% \quad (2)$$

High precision indicates that the model rarely misclassifies legitimate (ham) messages as spam, an essential factor in reducing false alarms.

4.2.3. Recall

Recall, also known as Sensitivity or True Positive Rate, measures the model's ability to correctly identify actual spam messages.

$$Recall = \frac{TP}{TP + FN} 100\% \quad (3)$$

A high recall score means the system captures most spam messages and minimizes the risk of letting harmful messages slip through as ham.

4.2.4. F1-Score

The F1-score is the harmonic mean of Precision and Recall, offering a balanced view of the model's performance. It is beneficial when the data is imbalanced.

$$F1 - Score = 2 \frac{(Precision)(Recall)}{(Precision) + (Recall)} \quad (4)$$

A high F1-score indicates the model maintains a good trade-off between catching spam and avoiding false alarms. Together, these metrics ensure a thorough and balanced evaluation of the proposed SMS spam detection framework across different performance dimensions [42].

4.2.5. AUC-ROC

We also report the Receiver Operating Characteristic (ROC) curve and its Area Under the Curve (AUC) score. The ROC illustrates the trade-off between true and false positive rates, while the AUC summarizes the

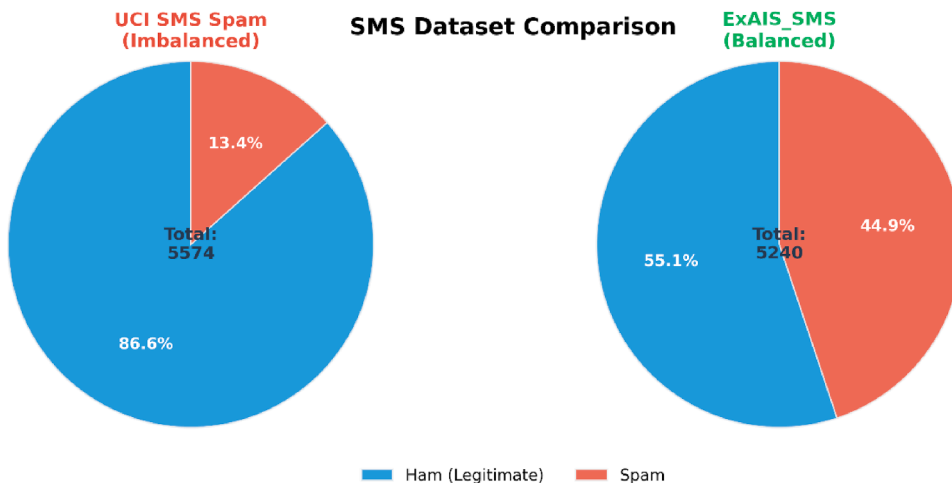


Fig. 2. Distribution of spam and ham messages in the UCI SMS Spam Collection v.1 and ExAIS SMS datasets.

model's overall discriminative ability [43]. An AUC close to 1.0 indicates strong separability between spam and ham.

4.3. Model hyperparameters

The proposed hybrid SMS spam detection model's performance is influenced by several carefully selected hyperparameters that govern the behavior of the fuzzy logic module and the BiLSTM deep learning architecture. These parameters were empirically optimized via a grid search on validation data from the SMS Spam Collection dataset.

4.3.1. Fuzzy input layer parameters and rule base

The fuzzy input layer transforms ambiguous linguistic terms into numerical membership scores. The following parameters are defined:

- **Membership Function Type:** Gaussian, selected for its smooth curves and ability to model gradual transitions between linguistic terms.
- **The number of Fuzzy Sets per Feature** is 3 (Low, Medium, High), providing balanced granularity for modeling vague linguistic cues such as urgency or promotion.
- **Rule Base Size:** Derived from frequent fuzzy co-occurrence patterns. For example, *IF (Urgency is High AND URL Presence is High), Spam likelihood = High*.
- **Fuzzy Aggregation Operator:** Min-Max, used to compute combined truth values across rules.

To enhance reproducibility and clarity, we explicitly provide an illustrative subset of the fuzzy rules applied in the LMA module. In theory, combining four linguistic variables (urgency, promotion, monetary incentive, and suspicious URL presence) with three fuzzy sets each (low, medium, high) could yield up to 81 rules. However, many combinations are redundant or semantically implausible. Therefore, in this study, we designed a compact, interpretable rule base comprising 23 fuzzy rules that capture the most frequent and meaningful co-occurrence patterns observed in the SMS datasets. Table 2 presents examples of fuzzy rules, and the complete rule list is in Appendix A (Table A1).

The complete rule base comprises 23 rules of similar structure, balancing comprehensiveness with interpretability. This design ensures the fuzzy module remains computationally efficient and transparent while effectively modeling linguistic ambiguity in SMS spam detection. All fuzzy rules in this study were manually designed based on observed linguistic patterns in the datasets, ensuring both interpretability and domain relevance.

4.3.2. BiLSTM deep learning parameters

The BiLSTM network learns temporal and contextual patterns from token sequences. Table 3 shows the hyperparameters.

4.3.3. Feature fusion & output layer

- **Fusion Strategy:** Concatenation of fuzzy and BiLSTM outputs.

Table 2

Example fuzzy rules applied in the LMA module.

Rule ID	Fuzzy Rule	Spam Likelihood
R1	IF Urgency is High AND URL Presence is High THEN Spam Likelihood = 0.92	0.92 (High)
R2	IF Promotion is Medium AND Monetary Incentive is High THEN Spam Likelihood = 0.87	0.87 (Medium-High)
R3	IF Suspicious Content Level is High AND Urgency is Medium THEN Spam Likelihood = 0.81	0.81 (Medium)
R4	IF Promotion is Low AND URL Presence is Low THEN Spam Likelihood = 0.10	0.10 (Low; likely ham)

Table 3

Model hyperparameters used in the proposed hybrid system's fuzzy logic and BiLSTM components.

Parameter	Value
Embedding Dimension	100
Embedding Type	Pre-trained Word2Vec (static)
BiLSTM Layers	1
Hidden Units per Layer	128
Dropout Rate	0.4
Activation Function	ReLU
Batch Size	64
Optimizer	Adam
Learning Rate	0.001
Loss Function	Binary Cross-Entropy
Epochs	15

- **Fully Connected Layer:** 1 hidden layer (64 units) followed by a sigmoid activation.
- **Output DimExtension:** 1 (binary classification: Spam vs. Ham).

Hyperparameters were selected via grid search on a held-out validation split of the UCI dataset, then fixed across both datasets to ensure a fair comparison.

5. Results and analysis

The proposed fuzzy-deep hybrid SMS spam detection model was evaluated on two benchmark datasets (UCI SMS Spam Collection v.1 and ExAIS_SMS), using widely accepted metrics: accuracy, precision, recall, and F1-score. The goal was to validate the model's predictive performance and interpretability, and to assess its adaptability across linguistic contexts.

5.1. UCI SMS spam collection dataset performance

Table 4 presents the proposed model's performance compared with existing methods on the UCI dataset. Our hybrid model achieved 99.40 % accuracy, outperforming both conventional and deep learning baselines. Notably, it maintained substantial precision (0.993) and recall (0.986), indicating its effectiveness in reducing false positives and false negatives.

These results demonstrate the robustness of integrating fuzzy reasoning with deep contextual modeling for SMS classification, particularly for noisy, obfuscated spam messages. Fig. 3 visually compares the proposed model's performance against other state-of-the-art approaches, emphasizing its superior accuracy and F1-score on the UCI dataset.

5.2. ExAIS_SMS dataset performance

As shown in Table 5, the proposed model also significantly outperforms traditional ML models on the ExAIS_SMS dataset, achieving

Table 4

Performance comparison of the proposed model and existing approaches on the UCI SMS Spam dataset.

Approach	Accuracy	Precision	Recall	F1-score
RF + TF-IDF [11]	97.50 %	0.98	0.97	0.98
J48, KNN, Decision Tree [12]	97.05 %	0.97	0.95	0.94
MLR + ELM [13]	98.20 %	0.90	0.98	0.94
Binary PSO + Fuzzy Rules [14]	98.50 %	0.93	0.98	0.95
ANN [18]	98.80 %	0.98	0.99	0.99
Modified Transformer [21]	98.92 %	0.97	0.94	0.96
CNN + GRU [20]	99.07 %	0.99	0.99	0.99
CNN + LSTM (parallel) [19]	99.28 %	0.99	0.98	0.99
BERT + Triple GCN [23]	99.28 %	0.99	0.97	0.98
Proposed method	99.40 %	0.993	0.986	0.989

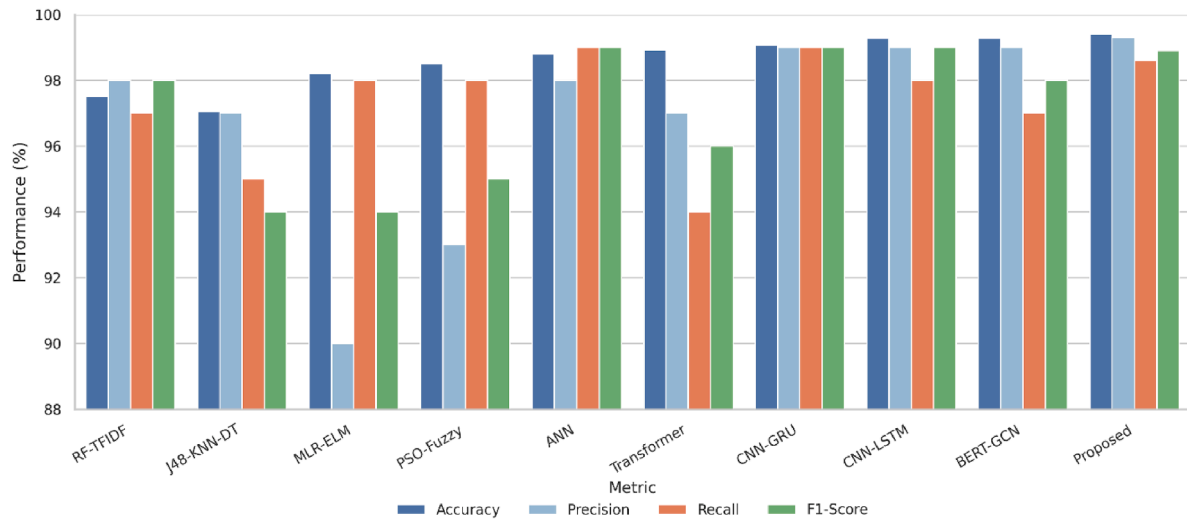


Fig. 3. Performance comparison of the methods on the UCI SMS Spam Collection v.1 dataset.

Table 5

Performance comparison of the proposed model and existing approaches on the ExAIS_SMS dataset.

Approach	Accuracy	Precision	Recall	F1-score
C4.5 [15]	60.65 %	0.71	0.60	0.57
Decision Table [15]	71.09 %	0.73	0.71	0.69
J48 [15]	73.63 %	0.78	0.73	0.71
BayesNet [15]	79.10 %	0.81	0.79	0.78
Naïve Bayes (NB) [15]	84.40 %	0.84	0.84	0.84
BiLSTM [22]	93.40 %	0.96	0.91	0.94
Proposed Method	95.8 %	0.968	0.959	0.964

95.8 % accuracy and a balanced F1-score of 0.964. This dataset presents a more linguistically diverse and balanced class distribution, offering a realistic environment to test the model's adaptability.

This indicates the model's strength in achieving high accuracy and maintaining a balance between sensitivity and precision, especially under different linguistic variations. Fig. 4 provides a visual comparison of model performance across various techniques on the ExAIS_SMS dataset, clearly highlighting the superiority of the proposed fuzzy-deep hybrid model in terms of precision, recall, and F1-score.

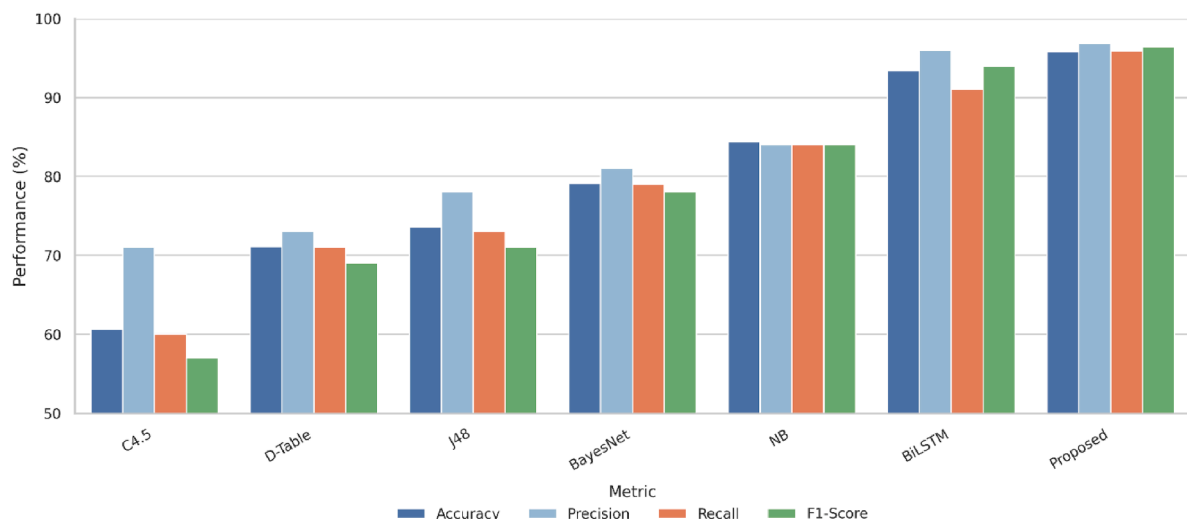


Fig. 4. Performance comparison of the methods on the ExAIS_SMS dataset.

5.3. ROC and AUC analysis

We report the ROC curves and the AUC. The ROC illustrates the trade-off between the true positive rate (TPR) and the false positive rate (FPR) (see Fig. 5), while the AUC summarizes overall discriminative ability. An AUC close to 1.0 indicates strong separability between spam and ham. Confidence intervals (95 % CI) were estimated via a 10,000-rep bootstrap over test folds.

5.4. Explainability performance

The novel part of the proposed model lies in its integrated Explainable AI (XAI) module, which enhances the interpretability of predictions without compromising performance. While many deep learning models achieve high accuracy, they often function as black boxes, making them unsuitable for deployment in environments that demand transparency, such as financial services, telecommunications, and regulatory-compliant systems [44]. Our model bridges this gap by providing token-level justifications for each classification decision, whether the message is labeled as spam or ham. The XAI component fuses two complementary mechanisms: (1) Attention-based attribution from the BiLSTM network, and (2) Fuzzy logic reasoning from linguistic membership scores. This dual-layer interpretability enables the model to

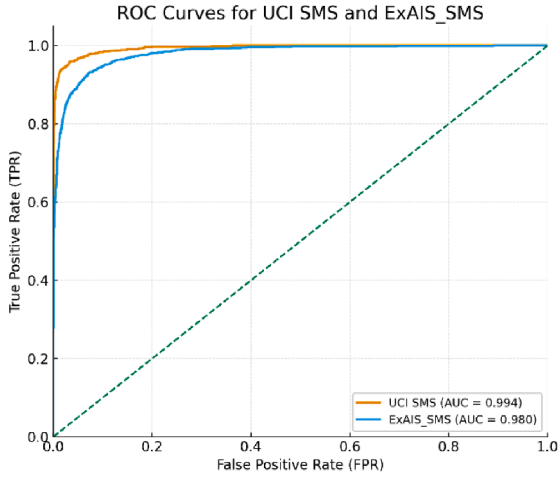


Fig. 5. Combined ROC curves for the UCI SMS Spam and ExAIS_SMS datasets, showing high separability with AUC values of 0.994 and 0.980, respectively.

Table 6

Token-level spam attribution for an example spam message.

Token	Spam Score (Fuzzy + DL)
Congratulations	0.91
You have	0.10
won	0.94
a prize	0.89
Click here	0.95

identify which tokens influenced the decision and explain *why*, in terms aligned with human reasoning.

5.4.1. Spam message explanation

Table 6 demonstrates the model's interpretability for a prototypical spam message from the UCI SMS Spam Collection: "Congratulations! You have won a prize. Click here to claim." Each token's spam contribution is quantified using the combined outputs of the fuzzy logic layer and the BiLSTM attention mechanism. High scores are assigned to promotional and call-to-action phrases, aligning with known spam indicators.

Tokens such as "won," "prize," and "click here" elicited strong activations in both the fuzzy membership functions (e.g., high promotion, urgency) and BiLSTM attention weights, yielding a transparent, interpretable spam classification. **Fig. 6** shows high attribution scores for key spam terms such as "won" and "click here," highlighting their strong influence on spam classification.

5.4.2. Ham message explanation

To illustrate the model's ability to distinguish non-spam messages with equal interpretability, consider the following benign message from the UCI SMS Spam Collection: "Are we still meeting at 7 pm for dinner?" As shown in **Table 7**, the model assigns low spam scores to all tokens, correctly identifying the message as legitimate communication. The message lacks promotional language, urgency cues, or obfuscated links. As such, neither the fuzzy rules nor the attention mechanism were activated significantly, demonstrating the model's reliability in classifying legitimate SMS. **Fig. 7** displays low attribution scores across all

tokens in a ham message, confirming the model's ability to identify non-spam content correctly.

We further examined a telecom-specific spam message to demonstrate applied relevance: "Your mobile account has been suspended. Click <http://bit.ly/xyz> to verify your details immediately." The model classified this instance as spam with high confidence (0.96). Attention highlighted key tokens such as "suspended" and "verify", while fuzzy rules activated the combination of High Urgency and Suspicious URL. The resulting explanation, "classified as spam because it conveys urgency and contains a suspicious link," mirrors the reasoning used by telecom analysts, underscoring the practicality of the proposed XAI module.

5.5. Regulatory implications of explainability

Beyond technical interpretability, the proposed XAI module has critical regulatory implications. Recent frameworks such as the European Union's AI Act and the General Data Protection Regulation (GDPR) emphasize transparency, accountability, and the "right to explanation" in automated decision-making [45]. In the context of SMS spam detection, token-level and rule-based justifications ensure that classifications can be audited and explained to users and regulators. This enhances user trust and positions the system as compliant with emerging legal requirements for fairness and accountability in AI-driven communication security.

5.6. Computational complexity

To evaluate the computational efficiency of the proposed fuzzy-deep hybrid model, experiments were conducted on a workstation equipped with an Intel Core i7 CPU, 16 GB of RAM, and an NVIDIA RTX 3090 GPU. The architecture comprises a BiLSTM with 128 hidden units and a lightweight fuzzy logic layer based on Gaussian membership functions. Empirical analysis shows that a single forward pass requires approximately 2.1–2.2 GFLOPs, with memory consumption ranging from 1.9 to 2.3 GB across datasets. On the UCI SMS Spam dataset, training completed in <4 min per epoch, and inference averaged 5.2 ms per message, while on the ExAIS_SMS dataset, inference averaged 6.8 ms per message. These results demonstrate that the model achieves high accuracy (99.40 % on UCI and 95.80 % on ExAIS_SMS) with modest computational cost, highlighting its practicality for real-time or near-real-time spam detection in standard workstation and server-grade environments. **Table 8** illustrates the Computational performance of the proposed method.

This interpretability capability represents a significant advancement over existing SMS spam detection systems, which typically offer little to no explanation of their predictions. The proposed XAI framework enhances user trust and regulatory auditability by providing granular,

Table 7

Token-level spam attribution for an example ham message.

Token	Spam Score (Fuzzy + DL)
Are	0.05
we	0.04
still	0.06
meeting	0.07
for dinner	0.08

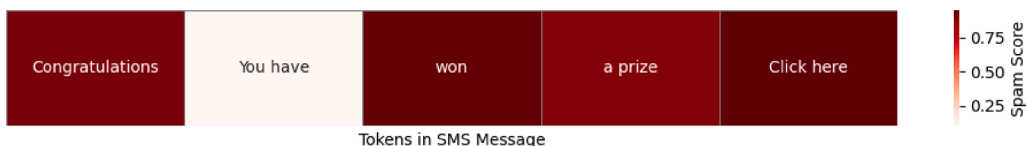


Fig. 6. Heatmap of token-level spam attribution for a spam message.

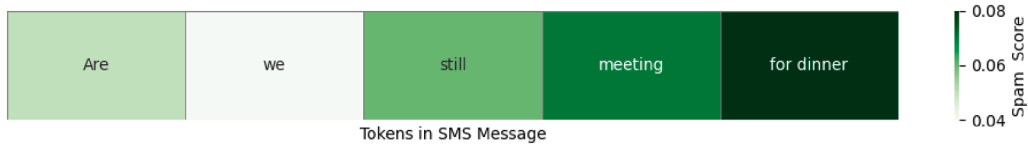


Fig. 7. Heatmap of token-level spam attribution for a ham message.

Table 8

Computational Performance on UCI and ExAIS_SMS Datasets.

Dataset	Accuracy (%)	Inference Time (ms/msg)	Approx. FLOPs (G)	Memory Use (GB)
UCI SMS Spam	99.40	5.2	2.2	2.3
ExAIS_SMS	95.80	6.8	2.1	1.9

rule-based, and attention-aware insights. It supports error analysis, system debugging, and user-specific customization of spam filters. In practical terms, the model transforms spam detection from a binary outcome into a transparent, explainable decision-making process, an essential requirement for AI adoption in high-stakes domains. While the proposed model demonstrates strong performance and interpretability, several limitations should be considered. The first limitation is that the current study focuses only on English datasets. Future work will explore multilingual adaptation strategies, such as leveraging cross-lingual embeddings and transfer learning, to extend applicability across different languages and dialects. Second, the model is designed for binary classification (spam vs. ham) and may not generalize well to more granular categories such as promotional, transactional, or phishing-specific messages. Third, although the XAI module offers meaningful token-level insights, its current format focuses on individual message interpretation and does not yet support longitudinal analysis or user-behavior-aware explanations. Future work can address these aspects to enhance generalizability and contextual awareness.

6. Conclusion

This study proposed a fuzzy-deep hybrid framework for SMS spam detection that integrates fuzzy logic with a bidirectional LSTM (BiLSTM) backbone to achieve high predictive accuracy and interpretability. The fuzzy module employs Gaussian membership functions to capture linguistic vagueness, while the BiLSTM effectively models sequential dependencies within SMS content. An Explainable AI module provides

token-level attention and rule-based explanations to enhance transparency, ensuring that classification outcomes are traceable and trustworthy. The framework was rigorously evaluated on two benchmark datasets, the UCI SMS Spam Collection and the ExAIS_SMS corpus, covering widely recognized and linguistically diverse real-world scenarios. Results confirmed the effectiveness of the proposed approach, with accuracies of 99.40 % on UCI and 95.80 % on ExAIS_SMS, supported by substantial precision, recall, F1-scores, and AUC values of 0.994 (UCI) and 0.980 (ExAIS_SMS). Beyond predictive performance, the model demonstrated computational efficiency, requiring <2.5 GB of memory and averaging under 7 ms per inference on a consumer-grade GPU, confirming its scalability for real-time telecom deployment. Including interpretable explanations further aligns the system with emerging regulatory requirements, such as the GDPR and the EU AI Act, thereby reinforcing its practical relevance. Nonetheless, limitations remain in the language scope and binary-class focus, motivating future work on multilingual adaptation, fine-grained spam categorization, and extended explainability. Overall, the proposed framework delivers a balanced solution that combines accuracy, interpretability, and efficiency, making it a reliable contribution to real-world spam detection systems.

CRedit authorship contribution statement

Ali kadhim Jasim: Writing – original draft, Data curation, Conceptualization. **Fuqdan A. Al-Ibraheemi:** Writing – review & editing, Visualization. **Hussein Alaa Alkaabi:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Fuzzy Rule Base

The LMA module uses a compact fuzzy rule base to enhance interpretability, modeling four linguistic variables Urgency, Promotion, Monetary Incentive, and URL Presence/Suspicious Content each with Low/Medium/High Gaussian memberships. IF-THEN rules produce a defuzzified Spam Likelihood $\in [0,1]$, which is fused with the BiLSTM score; the rule set focuses on frequent, semantically meaningful co-occurrences while avoiding redundancy. Table A1 shows the used rules.

Table A1

Comprehensive fuzzy rule base for the LMA module.

Rule ID	Fuzzy Rule (IF ... THEN ...)	Spam Likelihood
R1	IF Urgency is High AND URL Presence is High THEN Spam Likelihood = 0.92 (High)	0.92
R2	IF Promotion is Medium AND Monetary Incentive is High THEN Spam Likelihood = 0.87 (Med-High)	0.87
R3	IF Suspicious Content Level is High AND Urgency is Medium THEN Spam Likelihood = 0.81 (Medium)	0.81
R4	IF Promotion is Low AND URL Presence is Low THEN Spam Likelihood = 0.10 (Low; ham)	0.10
R5	IF Urgency is Medium AND Promotion is High THEN Spam Likelihood = 0.75 (Med-High)	0.75
R6	IF Urgency is High AND Promotion is High THEN Spam Likelihood = 0.93 (High)	0.93
R7	IF Monetary Incentive is High AND URL Presence is Medium THEN Spam Likelihood = 0.84 (High)	0.84
R8	IF Promotion is High AND Suspicious Content Level is High THEN Spam Likelihood = 0.95 (High)	0.95
R9	IF Urgency is Low AND Promotion is Low THEN Spam Likelihood = 0.15 (Low; ham)	0.15
R10	IF Urgency is High AND Monetary Incentive is High THEN Spam Likelihood = 0.91 (High)	0.91

(continued on next page)

Table A1 (continued)

Rule ID	Fuzzy Rule (IF ... THEN ...)	Spam Likelihood
R11	IF Promotion is Medium AND URL Presence is Medium THEN Spam Likelihood = 0.70 (Medium)	0.70
R12	IF Urgency is Medium AND Suspicious Content Level is Medium THEN Spam Likelihood = 0.65 (Medium)	0.65
R13	IF Promotion is High AND URL Presence is High THEN Spam Likelihood = 0.94 (High)	0.94
R14	IF Urgency is High AND Suspicious Content Level is High THEN Spam Likelihood = 0.96 (Very High)	0.96
R15	IF Promotion is Low AND Monetary Incentive is Low THEN Spam Likelihood = 0.20 (Low; ham)	0.20
R16	IF Urgency is Medium AND Promotion is Medium THEN Spam Likelihood = 0.55 (Medium)	0.55
R17	IF Monetary Incentive is Medium AND URL Presence is High THEN Spam Likelihood = 0.78 (Med-High)	0.78
R18	IF Promotion is High AND Monetary Incentive is High THEN Spam Likelihood = 0.97 (Very High)	0.97
R19	IF Urgency is Low AND URL Presence is Medium THEN Spam Likelihood = 0.40 (Low-Med)	0.40
R20	IF Suspicious Content Level is Medium AND Promotion is High THEN Spam Likelihood = 0.72 (Med-High)	0.72
R21	IF Urgency is Medium AND URL Presence is Low THEN Spam Likelihood = 0.35 (Low-Med; ham-leaning)	0.35
R22	IF Urgency is High AND Promotion is High AND URL Presence is High THEN Spam Likelihood = 0.98 (Very High)	0.98
R23	IF All Features are Low (Urgency, Promotion, Incentive, URL) THEN Spam Likelihood = 0.05 (Very Low; ham)	0.05

References

[1] L.N. Lota, B.M. Hossain, A systematic literature review on sms spam detection techniques, *Int. J. Inf. Technol. Comput. Sci. (IJITCS)* 9 (7) (2017) 42–50.

[2] M. Salman, M. Ikram, M.A. Kaafar, Investigating evasive techniques in sms spam filtering: a comparative analysis of machine learning models, *IEEe Access*. 12 (2024) 24306–24324.

[3] H.A. Al-Kabbi, M.-R. Feizi-Derakhshi, S. Pashazadeh, Multi-type feature extraction and early fusion framework for SMS spam detection, *IEEe Access*. 11 (2023) 123756–123765, <https://doi.org/10.1109/ACCESS.2023.3327897>.

[4] SlickText, 17 Spam Text Statistics For 2025 & Spam Text Messages Examples, SlickText Blog, Mar. 6, 2025 [Online]. Available: <https://www.slicktext.com/blog/2022/10/17-spam-text-statistics-for-2022/> [Accessed: Aug. 30, 2025].

[5] A.A. Akinyelu, Advances in spam detection for email spam, web spam, social network spam, and review spam: ML-based and nature-inspired-based techniques, *J. Comput. Secur.* 29 (5) (2021) 473–529.

[6] H. Al-Kaabi, A.D. Darroudi, A.K. Jasim, Survey of SMS spam detection techniques: a taxonomy, *AlKadhim J. Comput. Sci.* 2 (4) (2024) 23–34.

[7] V. Buhrmester, D. Münch, M. Arens, Analysis of explainers of black box deep neural networks for computer vision: a survey, *Mach. Learn. Knowl. Extr.* 3 (4) (2021) 966–989.

[8] M.M.A. Busst, K.S.M. Anbananthen, S. Kannan, J. Krishnan, S. Subbiah, Ensemble BiLSTM: a novel approach for aspect extraction from online text, *IEEe Access*. 12 (2024) 3528–3539.

[9] Z. Hemmat, M. Mehraeen, R.A. Fattahi, F. Shirani, Identifying the design requirements of explainable artificial intelligence systems, *J. Strateg. Manag. Stud.* 15 (59) (2024) 45–67.

[10] M. Foozy, C. Feresa, R. Ahmad, M.F. Abdollah, A practical rule-based technique for splitting SMS phishing from SMS spam for better accuracy on mobile devices, *Int. Rev. Comput. Softw.* 9 (10) (2014) 1776–1782.

[11] N.N.A. Sjarif, N.F.M. Azmi, S. Chuprat, H.M. Sarkan, Y. Yahya, S.M. Sam, SMS spam message detection using term frequency-inverse document frequency and random forest algorithm, *Procedia Comput. Sci.* 161 (2019) 509–515.

[12] V.A. Saeed, A method for SMS spam message detection using machine learning, *Artif. Intell. Robot. Dev. J* 3 (2023) 214–228.

[13] Z.H. Ali, H.M. Salman, A.H. Harif, SMS spam detection using multiple linear regression and extreme learning machines, *Iraqi J. Sci.* (2023) 6342–6351.

[14] S.M. Hameed, Z.H. Ali, SMS spam detection based on fuzzy rules and binary particle swarm optimization, *Int. J. Intell. Eng. Syst.* 14 (2) (2021).

[15] Sani, M.M., Abdulrahman, S.M., & Adamu, A. (2025). Improving SMS spam detection using classical machine learning algorithms.

[16] M. Giannakos, R. Azevedo, P. Brusilovsky, M. Cukurova, Y. Dimitriadis, D. Hernandez-Leo, B. Rienties, The promise and challenges of generative AI in education, *Behav. Inf. Technol.* 44 (11) (2025) 2518–2544.

[17] A.N. Kamber, H. Alkaabi, M. Al-Rekabi, A.K. Jasim, Explainable AI for medical imaging: a taxonomy based on clinical task requirements, *PERFECT: J. Smart Algorithms* 2 (2) (2025) 72–77.

[18] S. Sheikhi, M.T. Kheirabadi, A. Bazzazi, An effective model for SMS spam detection using content-based features and an averaged neural network, *Int. J. Eng.* 33 (2) (2020) 221–228.

[19] M.R.F. Derakhshi, E. Zafarani-Moattar, H.A.A. Al-Kabi, A.H.J. Almarashy, Pclif: parallel cnn-lstm fusion model for sms spam filtering, in: *BIO Web of Conferences* 97, EDP Sciences, 2024 00136.

[20] H.C. Altunay, Z. Albayrak, SMS spam detection system based on deep learning architectures for turkish and english messages, *Appl. Sci.* 14 (24) (2024) 11804.

[21] X. Liu, H. Lu, A. Nayak, A spam transformer model for SMS spam detection, *IEEe Access* 9 (2021) 80253–80263.

[22] O. Abayomi-Alli, S. Misra, A. Abayomi-Alli, A deep learning method for automatic SMS spam classification: performance of learning algorithms on indigenous dataset, *Concurr. Comput.: Pract. Exp.* 34 (17) (2022) e989.

[23] L. Shen, Y. Wang, Z. Li, W. Ma, SMS spam detection using BERT and multi-graph convolutional networks, *Int. J. Int. Net.* (2025).

[24] B.S. Aparna, S. Remya, M.J. Pillai, S.R. Subbareddy, Y.Y. Cho, ALBERT-BiLSTM cross-attention network with progressive knowledge distillation for multi-domain SMS spam classification, *Results Eng.* (2025) 106727.

[25] Alkaabi, H., Ibraheemi, F., Jasim, A., Idan, Z.S.I., & Alhelal, A.R. (2025). Arabic sms spam detection using arabert and dual feature extraction: a study on modern standard and iraqi dialects. <https://doi.org/10.21203/rs.3.rs-6832100/v1>.

[26] R. Khorashadizade, S. Jafarali Jassbi, A. Yari, Persian sms spam detection using machine learning and deep learning techniques, *Int. J. Web Res.* 5 (1) (2022) 56–65.

[27] I.S. Mambina, J.D. Ndibwile, D. Uwimpuhwe, K.F. Michael, Uncovering sms spam in swahili text using deep learning approaches, *IEEe Access* 12 (2024) 25164–25175.

[28] J. Gui, Y. Zhou, K. Yu, X. Wu, PSC-BERT: a spam identification and classification algorithm via prompt learning and spell check, *Knowl. Based. Syst.* 301 (2024) 112266.

[29] H.J. Aleqabie, M.S. Sfoq, R.A. Albeer, E.H. Abd, A review of TextMining techniques: trends, and applications in various domains, *Iraqi J. Comput. Sci. Math.* 5 (1) (2024) 9.

[30] A.A. Nafea, M.S. Muayad, R.R. Majeed, A. Ali, O.M. Bashaddadh, M.A. Khalaf, A. Steiti, A brief review on preprocessing text in arabic language dataset: techniques and challenges, *Babylon. J. Artif. Intell.* 2024 (2024) 46–53.

[31] B.I. Kzar, H.H. Safi, Systematic review of sentiment analysis and predict sarcastic, *J. Al-Qadisiyah Comput. Sci. Math.* 15 (2) (2023) 166. Page-.

[32] L.A. Rosewelt, D.N. Raju, E. Sujatha, A new sentiment and fuzzy aware product recommendation system using weighted Aquila optimization and GRNN in e-commerce, *Inf. Technol. Control* 52 (3) (2023) 617–637.

[33] K. Misra, J.T. Rayz, Finding fuzziness in neural network models of language processing, in: *North American Fuzzy Information Processing Society Annual Conference*, Springer International Publishing, Cham, 2021, June, pp. 278–290.

[34] H.A. Al-Kabbi, M.R. Feizi-Derakhshi, S. Pashazadeh, A hierarchical two-level feature fusion approach for SMS spam filtering, *Intell. Autom. Soft Comput.* 39 (4) (2024).

[35] J.N. Mogan, C.P. Lee, K.M. Lim, Ensemble CNN-ViT using feature-level fusion for gait recognition, *IEEe Access* (2024).

[36] M. Umer, Z. Imtiaz, M. Ahmad, M. Nappi, C. Medaglia, G.S. Choi, A. Mehmood, Impact of convolutional neural network and FastText embedding on text classification, *Multimed. Tools. Appl.* 82 (4) (2023) 5569–5585.

[37] Doshi-Velez, F., & Kim, B. (2017). Towards a rigorous science of interpretable machine learning. *arXiv prepr. arXiv:1702.08608*.

[38] M. Liu, H. Zhang, Z. Xu, K. Ding, The fusion of fuzzy theories and natural language processing: a state-of-the-art survey, *Appl. Soft. Comput.* (2024) 111818.

[39] T.A. Almeida, J.M.G. Hidalgo, A. Yamakami, Contributions to the study of SMS spam filtering: new collection and results, in: *Proceedings of the 11th ACM symposium on Document engineering*, 2011, pp. 259–262.

[40] O. Abayomi-Alli, ExAIS SMS dataset, Accessed March 09, 2021, <https://github.com/AbayomiAlli/SMS-Spam-Dataset>, 2021.

[41] E.A. Saeed, A.D. Jasim, M.A.A. Malik, Cuneiform text dialect identification using machine learning algorithms and natural language processing (nlp), *Iraqi J. Inf. Commun. Technol.* 7 (2) (2024) 26–40.

[42] A.I. Jabbooree, L.M. Khanli, P. Salehpour, S. Pourbahrami, A novel facial expression recognition algorithm using geometry β -skeleton in fusion based on deep CNN, *Image Vis. Comput.* 134 (2023) 104677.

[43] S.S. Tiwari, H.R. Goyal, A. Shrivastava, K. Yadav, A.L.N. Rao, Clinical text analysis with natural language processing: a BERT-based approach, in: *2024 International Conference on Communication, Computer Sciences and Engineering (IC3SE)*, IEEE, 2024, May, pp. 1382–1388.

[44] M. Saarela, V. Podgorelec, Recent applications of Explainable AI (XAI): a systematic literature review, *Appl. Sci.* 14 (19) (2024) 8884.

[45] B. Goodman, S. Flaxman, European Union regulations on algorithmic decision-making and a “right to explanation, *AI Mag.* 38 (3) (2017) 50–57.